

Matthew Lam

✉ ma.lam@mail.utoronto.ca | 📞 +1 416-830-3476 | 🌐 ml-matthew-lam

EDUCATION

University of Toronto

BASc in Engineering Science + PEY Co-Op | GPA: 3.67

Toronto, Ontario

Sep. 2025 – Apr. 2030

ACADEMIC ENRICHMENT

Quantum School for Young Students

Aug. 2024

University of Waterloo Institute for Quantum Computing

- 1 of ~200 students selected from across Canada to attend lectures from quantum science researchers & study topics such as polarization qubits.

PROJECTS

Water Electrolysis & Ionic Concentration | IB Chemistry Investigation

Oct. 2024 – Feb. 2025

- Designed & ran a water electrolysis experiment to determine the effect of varying NaOH concentration on electric current & rate of H₂ production.

Mathematics of the Consonance of Musical Intervals | IB Extended Essay

Oct. 2023 – Sep. 2024

- Self-studied Fourier Analysis & researched models of musical consonance based on frequency ratio, beating & roughness, including W. A. Sethares's model from *Tuning, Timbre, Spectrum, Scale* (1999).
- Applied FFT in MATLAB to analyze frequency spectra & plotted consonance/dissonance curves.

Interpolation Methods for Image Resizing | IB Mathematics Exploration Project

Oct. 2024 – Jan. 2025

- Self-studied nearest neighbour, bilinear, bicubic & bicubic spline interpolation methods.
- Applied interpolation methods to an image in MATLAB & compared results visually & with mean squared error.

Bridge Design Project | Course project in CIV102 (Structures & Materials course)

Nov. 2025

- Collaborated in a team of 3 to design a matboard bridge capable of supporting a load of 580 N.
- Wrote Python code to compute & plot shear force & bending moment envelopes & capacities with Matplotlib.

Top 20 in Desmos Global Math Art Contest | Age 15-16 Category

Dec. 2022 – Jan. 2023

- Created winning artwork incorporating visual and musical components using graphs of mathematical equations.

EXPERIENCE & LEADERSHIP

Ensemble Leader, Clarinet Section Leader & Peer Mentor

Sep. 2023 – May 2025

Toronto French School Wind Ensemble

Toronto, Ontario

- Ensemble won Silver Award in B600 class (highest performance level) at 2024 & 2025 MusicFest Canada national festival.
- Coached & managed an ensemble of 70+ musicians & organized rehearsals & concerts.

Founder, Music Director & Musician

Jun. 2024 – Sep. 2024

Music for Mother Earth

Toronto, Ontario

- Founded a 7-member ensemble, designed & built wind instruments using locally grown vegetables & repurposed wood, & performed at a community fair with ~200 guests.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB | **Libraries:** Matplotlib | **Other Tools:** LaTeX

AWARDS & HONOURS

Governor General's Academic Medal

2025

Highest academic standing in high school graduating class.

Anna Giles Medal

2025

Awarded to one student in high school graduating class for demonstrating exceptional honour, generosity & compassion, as voted by peers & teachers.